BIOCHAR AND DUNG BEETLES

Adaptation
Mitigation
Productivity



Regeneration

Climate & Agricultural Support PTY LTD

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https://climateagriculturalsupport.com/



An opportunity to improve production, soil health, plant health, animal health and climate change







Fleurieu Milk Company

BIOCHAR
DAIRY FEEDING TRAIL

- 4 x dairy's of 250 cattle
  - 1 fed Biochar type 1 90ml/head/day but issues with other additives
  - 1 fed Soft Agriculture - Pty Ltd – Mara Seeds Pty Ltd 200gms/head/day
  - 2 dairy's without biochar
- Measured 6 weekly milk yield, protein, fat, ICCC compared with long term data across 4 dairies
- Manure pH, and other properties
- Soil health (burying biochar)

ECONOMIC BENEFIT OF FEEDING BIOCHAR

Biochar type	Inclusion rate/COW	Number of Cows	Daily rate	Daily rate	Monthly rate	12month	Cost	Total Cost 12 months
Biochar cost	150gms	250	37500 gm	37.5 kg	1 tonne	12 tonne	$600/tonne	$7200



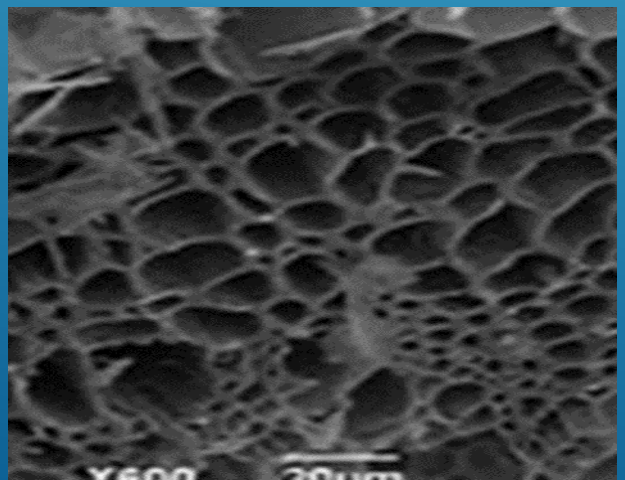
SA Milk and Solids	inc/head/ day	inc/day x 250 cows	$/day x 250 cows	$ inc/month	12months
49c/litre milk	1.5 litres/day	375l litres/day	$183/day	$5490	$65880

Also savings on fodder of roughly 2 round bales a week.
Estimate = $12,000
Plus profit inc of $58680
Improvement of $70680

OUTCOMES DAIRY FEEDING TRAIL

- Dairy Feeding Trial
  - Improved milk yield by around 1.5 litres/head/day
  - Reduced need for additional fodder
- The biochar is not digested and ends up in the dung!
- What happens in the Soil?







TALKING SH_T

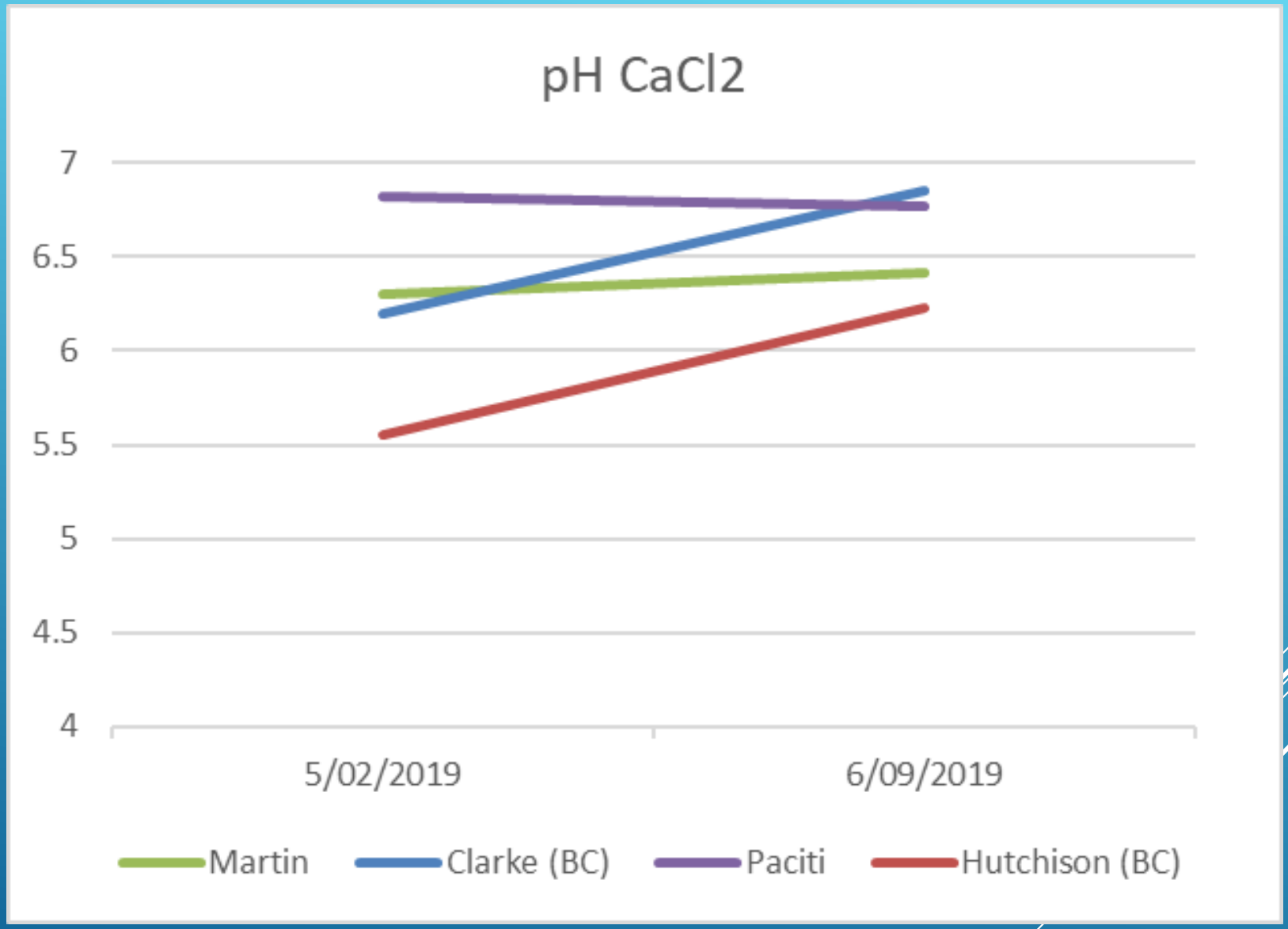
- 50 cows = 1 tonne of dung a day
- 250 cows = 5 tonne of dung in a day and 1825 tonnes in a year
- 1 dung = approx. 2 litres
- I dung covers approx. 1/10 square meter
- 1 cow can do 10-12 dungs a day covering 1 square meters each
- 250 cows cover 250 square meters or .025ha in 1 day
- 9.125ha in 1 year covered in 1 year
- Contains 1825 tonnes of dung inc 12 tonnes of biochar
- Over a year 1825 tonnes of dung buried containing 12 tonnes of biochar spread across 9ha
- It builds and spreads below and grows microbes etc
- Or Spread biochar and it will be buried.

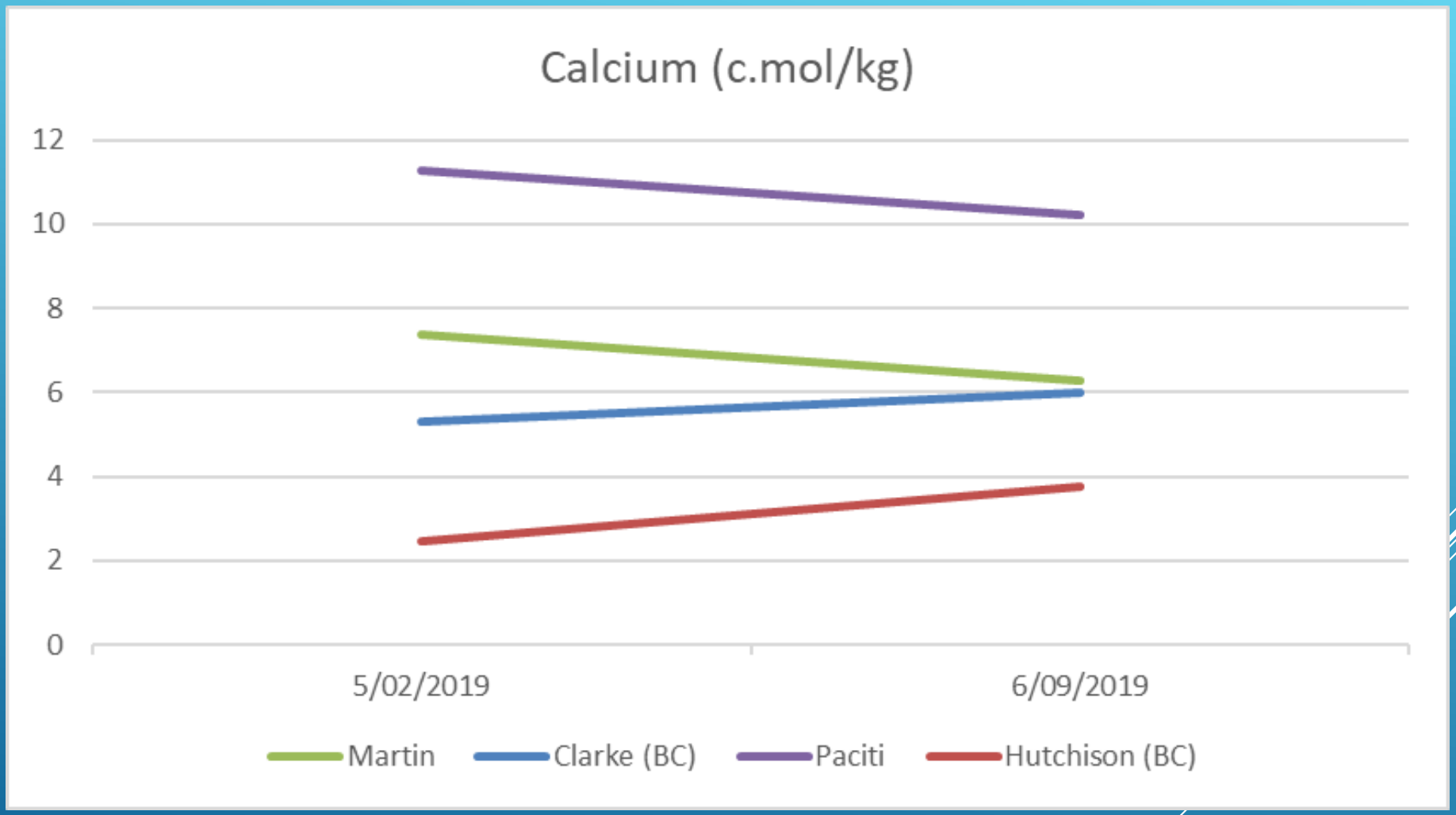


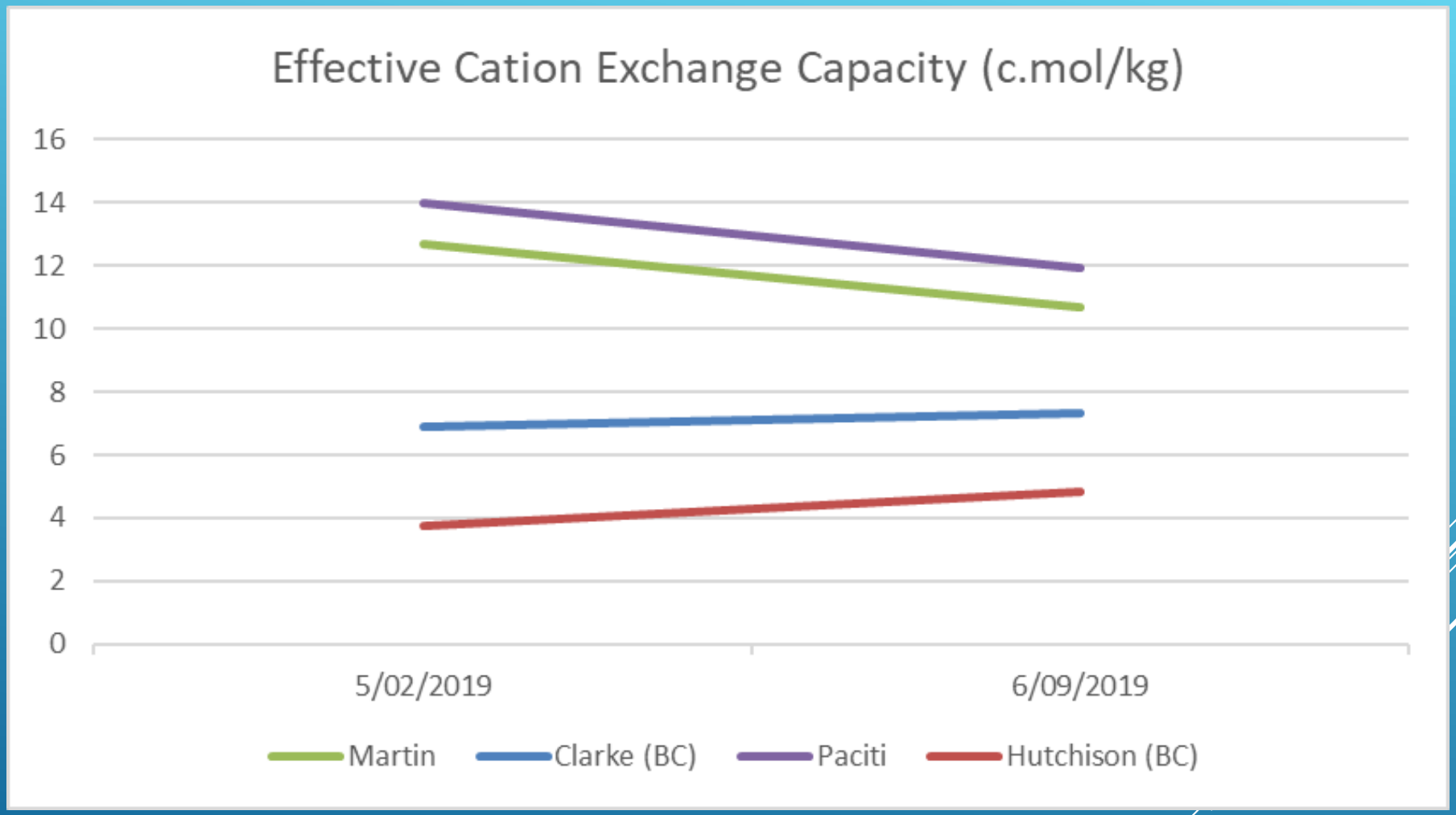


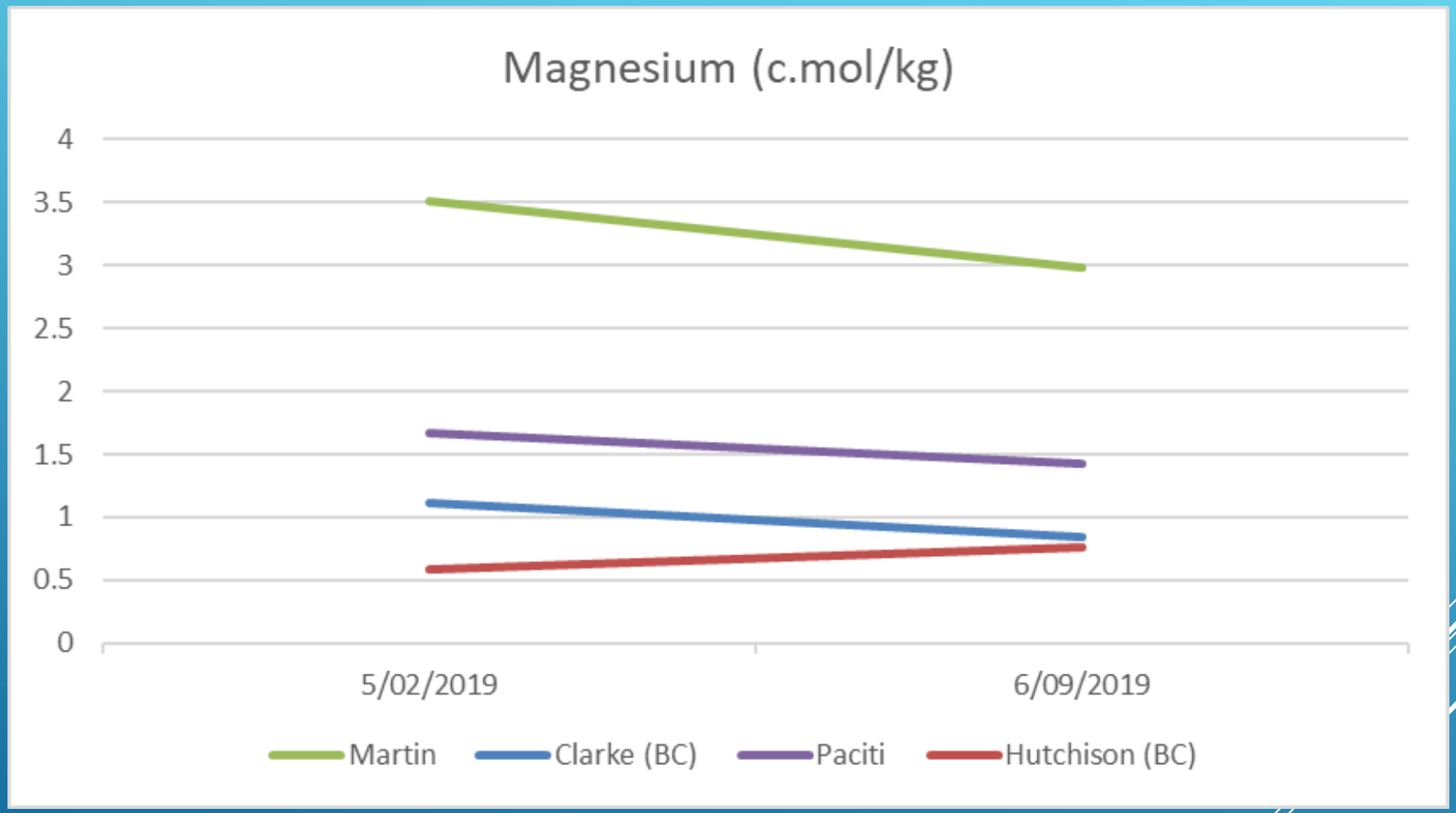




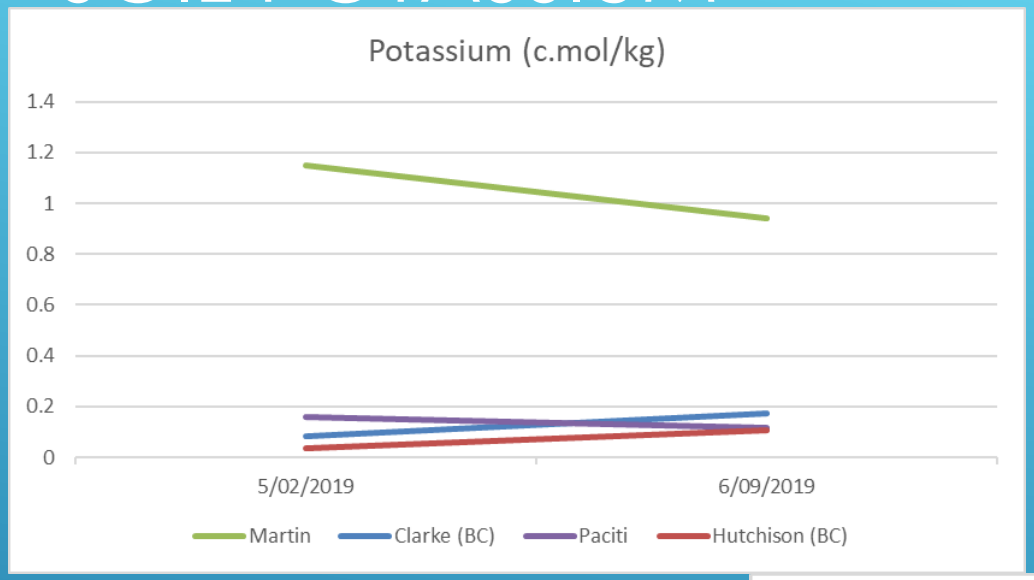

pH CaCl2
7
6.5
6
5.5
5
4.5
4
5/02/2019
6/09/2019
Martin
Clarke (BC)
Paciti
Hutchison (BC)

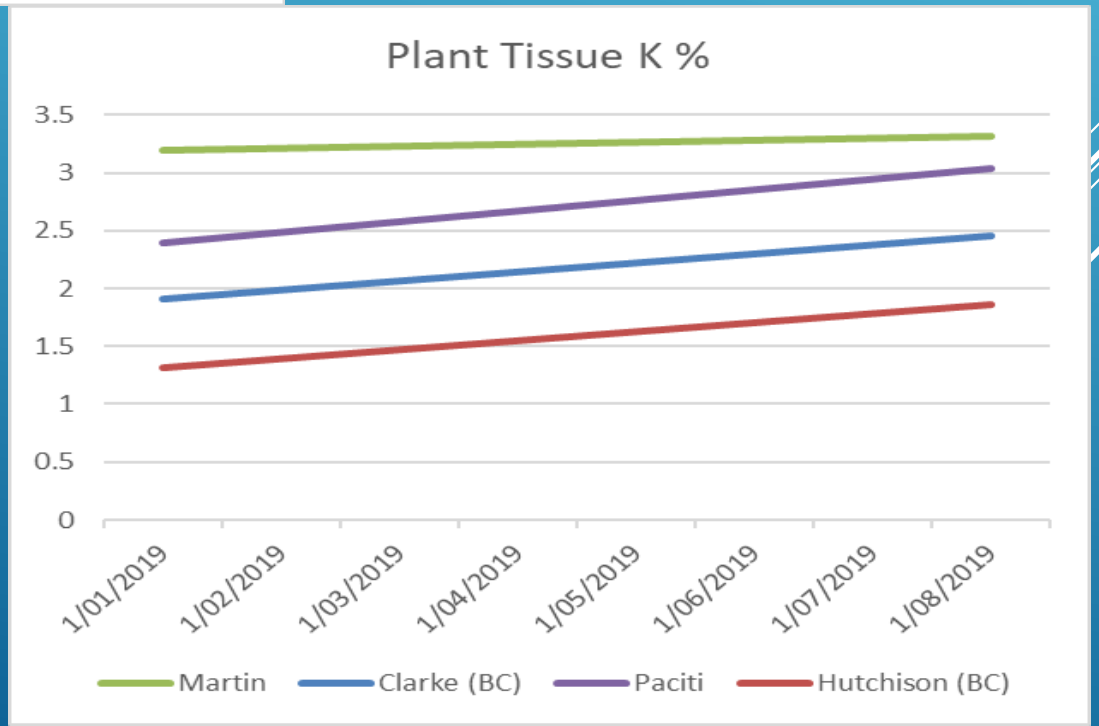

Calcium (c.mol/kg)
12
10
8
6
4
2
0
5/02/2019
6/09/2019
Martin
Clarke (BC)
Paciti
Hutchison (BC)


Effective Cation Exchange Capacity (c.mol/kg)
16
14
12
10
8
6
4
2
0
5/02/2019
6/09/2019
Martin
Clarke (BC)
Paciti
Hutchison (BC)


Magnesium (c.mol/kg)
4
3.5
3
2.5
2
1.5
1
0.5
0
5/02/2019
6/09/2019
Martin
Clarke (BC)
Paciti
Hutchison (BC)

SOIL POTASSIUM





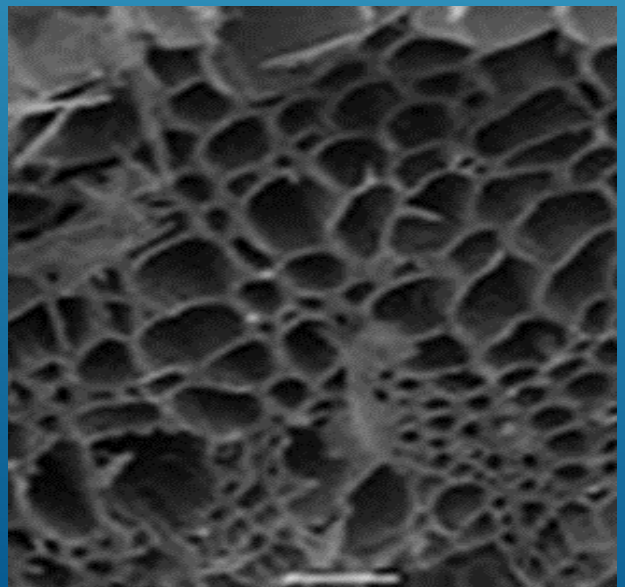
BIOCHAR AND DUNG BEETLE BURIAL













BENEFITS OF DUNG BEETLES

- Carbon burial at 20 to 60 cm depth
- Improve soil nutrition
- Reduce pasture fouling
- Reduce soil compaction
- Increase water infiltration
- Reduced soil erosion from water run-off
- Interrupt the life cycle of intestinal parasites
- Increase earthworms and microbial activity.



Public benefits

- Reduce carbon dioxide and nutrients (and sometimes pathogens) contaminating water bodies and the atmosphere;
- Reducing bush flies

Slde by Greg Dalton – Creation Care

Pottential dry matter t/ha		
t/ha	10%	20%
4	0.4	0.8
6	0.6	1.2
8	0.8	1.6

$ value/ha @ $150/tonne		
t/ha	10%	20%
4	$60	$120
6	$90	$180
8	$120	$240

$ value/100 hectares		
t/ha	10%	20%
4	$6,000	$12,000
6	$9,000	$18,000
8	$12,000	$24,000

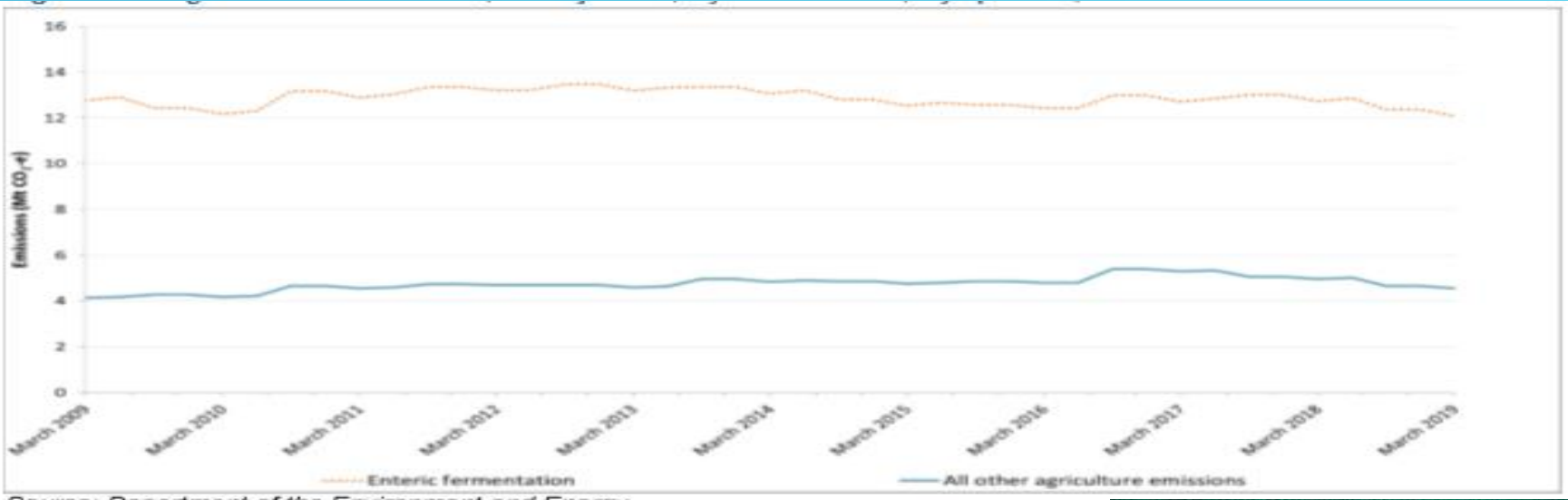
Site suitability Project

Defining site recommendations on the Fleurieu Peninsula for *O. Vacca*, *B. Bubalus* and *C. hispanis*



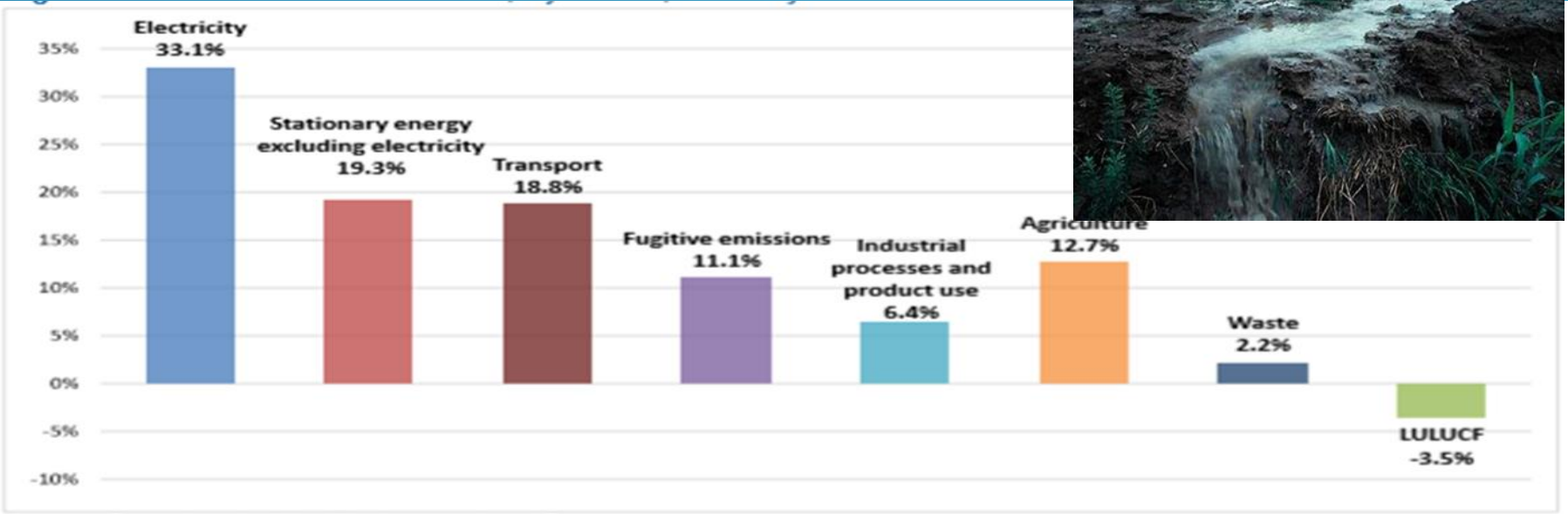


ENTERIC FERMENTATION FROM AGRICULTURE VS OTHER



Source: Department of the Environment and Energy





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